



ABERDARE INSTITUTE OF TECHNOLOGY

ELECTRICAL AND ELECTRONICS ENGINEERING

SEMICONDUCTOR THEORY EXAMINATION

Time: 3 Hours

Total Marks: 100

Instructions

1. Answer **ALL** questions.
2. Show all workings where necessary.
3. Draw neat and well-labeled diagrams where applicable.
4. The paper carries a total of **100 marks**.

Question 1 (20 Marks)

- a) Define the term **semiconductor**. (2 Marks)
- b) State any four characteristics of semiconductor materials. (4 Marks)
- c) Explain the semiconductor theory in relation to electrical conduction. (4 Marks)
- d) Distinguish between conductors, insulators and semiconductors. (6 Marks)
- e) State two practical applications of semiconductors. (4 Marks)

Question 2 (15 Marks)

- a) Describe the structure of matter, indicating the following:
 - i. Atom (1 Mark)
 - ii. Nucleus (1 Mark)
 - iii. Proton (1 Mark)
 - iv. Neutron (1 Mark)
 - v. Electron (1 Mark)
- b) Explain how electrons contribute to electrical conduction in conductors. (4 Marks)

- c) Explain how electrical conduction takes place in semiconductors. (4 Marks)
- d) State the difference between free electrons and valence electrons. (2 Marks)

Question 3 (15 Marks)

- a) Name two commonly used semiconductor materials. (2 Marks)
- b) Describe the properties of Silicon. (4 Marks)
- c) Describe the properties of Germanium. (4 Marks)
- d) Compare Silicon and Germanium under the following headings:
 - i. Cost (1 Mark)
 - ii. Temperature stability (1 Mark)
 - iii. Leakage current (1 Mark)
 - iv. Industrial applications (2 Marks)

Question 4 (15 Marks)

- a) Explain the process of doping in semiconductors. (3 Marks)
- b) Define:
 - i. P-type semiconductor material (2 Marks)
 - ii. N-type semiconductor material (2 Marks)
- c) Name two trivalent impurities used in P-type semiconductors. (2 Marks)
- d) Name two pentavalent impurities used in N-type semiconductors. (2 Marks)
- e) State the majority and minority charge carriers in:
 - i. P-type materials (2 Marks)
 - ii. N-type materials (2 Marks)

Question 5 (15 Marks)

- a) Draw and label a P-N junction diode. (3 Marks)
- b) Explain the formation of a depletion layer in a P-N junction. (4 Marks)
- c) Describe the operation of a diode under:
 - i. Forward biasing (4 Marks)
 - ii. Reverse biasing (4 Marks)

Question 6 (10 Marks)

- a) Define a transistor. (2 Marks)
- b) State three functions of a transistor in electronic circuits. (3 Marks)
- c) Draw and label the symbol of:
 - i. NPN transistor (2 Marks)
 - ii. PNP transistor (2 Marks)
- d) State one difference between NPN and PNP transistors. (1 Mark)

Question 7 (10 Marks)

- a) Explain the role of semiconductor devices in modern electronics. (4 Marks)
- b) State six applications of semiconductor devices. (6 Marks)

Question 8 (10 Marks)

- a) What is a hole in semiconductor theory? (2 Marks)
- b) State the function of a rectifier diode. (2 Marks)
- c) Explain why Silicon is preferred to Germanium in most electronic devices. (3 Marks)
- d) State three advantages of semiconductor devices over vacuum tubes. (3 Marks)